

SIMATS ENGINEERING
ASSESSMENT TOOL 2— ANNEXURE A
ARTICLE REVIEW — Course: Artificial Intelligence (CSA1711) | CO4

Selected Article

Zhao, L., Lee, S., & Jeong, S.-P. (2021). Decision Tree Application to Classification Problems with Boosting Algorithm. *Electronics*, 10(16), Article 1903. <https://doi.org/10.3390/electronics10161903>

Task 1: Article Summary

(a) Citation, Domain and ML Problem

Full citation: Zhao, L., Lee, S., & Jeong, S.-P. (2021). Decision Tree Application to Classification Problems with Boosting Algorithm. *Electronics*, 10(16), 1903. DOI: 10.3390/electronics10161903.

Domain / application area: Banking and financial services — direct-marketing analytics for a Portuguese retail bank.

ML problem addressed: Binary classification of whether a bank customer will subscribe to a term (fixed) deposit, based on demographic, contact-history, and campaign attributes. The target class distribution is heavily imbalanced, since far fewer customers subscribe than decline.

(b) Objective and Research Gap (150–200 words)

The authors set out to determine whether combining classical decision-tree induction with an ensemble boosting procedure can improve the reliability of customer-subscription prediction for a bank's telemarketing campaign. Earlier decision-tree studies applied C5.0 or CART largely as standalone classifiers and reported headline accuracy figures, but the authors observed that such single-tree models tend to overfit and, more importantly, perform poorly on the minority "subscriber" class once the dataset is imbalanced — a condition typical of real banking data. This is the research gap: prior work rarely examined how boosting and cost-sensitive learning jointly affect precision, sensitivity, and F1-score on the minority class rather than on overall accuracy alone. To close this gap, the authors fit C5.0 and CART trees with and without an AdaBoost-style boosting wrapper and, for C5.0, also introduce a cost matrix that penalises misclassifying an actual subscriber more heavily. They then compare all five variants using confusion-matrix metrics and 10-fold cross-validation, rather than accuracy in isolation, to identify which configuration best serves the bank's practical goal of identifying likely subscribers.

Task 2: Methodology Analysis

(a) ML Technique(s) and Dataset

The study builds two decision-tree variants — C5.0, which splits on information-gain ratio, and CART, which splits on the Gini coefficient — and strengthens each with an AdaBoost-style boosting procedure that trains a sequence of weak tree learners, re-weights misclassified samples after every round, and combines the weak learners into one strong classifier through a weighted vote. For C5.0 specifically, the authors also add a cost matrix that assigns a higher penalty to false negatives (missed subscribers), producing a fourth variant, C5.0 + CM.

The dataset is the well-known Bank of Portugal telemarketing dataset: 45,211 customer records in total, split into 40,690 samples used for model fitting and 4,521 held out for testing. Each record has 16 predictor attributes (7 continuous — age, balance, day, duration, campaign, pdays, previous — and 9 categorical, such as job, marital status, education, and outcome of the previous campaign) plus a binary target indicating term-deposit subscription. Before modelling, the authors checked Pearson correlation coefficients and variance-inflation factors (VIF) among the continuous variables; because all pairwise correlations were below 0.5 and every VIF was under 2, they concluded multicollinearity was not a concern and dimensionality reduction (e.g., PCA) was unnecessary.

(b) Mapping to CO4 and Methodological Strength/Limitation

The paper maps directly onto the Decision Trees strand of CO4: it applies two canonical decision-tree induction algorithms (C5.0 and CART) and layers an ensemble/boosting mechanism on top, which also connects to the broader idea of combining weak learners — a concept closely related to reinforcement-style iterative weight updates covered under CO4.

Strength: The use of a confusion matrix together with 10-fold cross-validation, rather than a single train/test accuracy figure, gives a more trustworthy estimate of generalisation performance and exposes how each model behaves specifically on the minority (subscriber) class.

Limitation: The authors rely on a single, moderately old public dataset and do not compare against non-tree baselines (e.g., logistic regression, SVM, or neural networks), so it is unclear whether the reported gains are specific to tree-based boosting or would also be matched by simpler statistical learners on the same data.

Task 3: Results & Critical Evaluation

(a) Key Results (Comparative Table)

The table below reports the test-set metrics for "positive = yes" (i.e., the customer subscribes), as presented by the authors.

Metric	C5.0	C5.0 + CM	C5.0 + Boosting	CART	CART + Boosting
Accuracy	0.9056	0.8960	0.9033	0.9031	0.9095
Precision	0.4560	0.7032	0.4518	0.3724	0.6523
Sensitivity (Recall)	0.6324	0.5431	0.6192	0.6502	0.4858
Specificity	0.9305	0.9591	0.9299	0.9213	0.9657
F1-score	0.5305	0.6129	0.5224	0.4736	0.5569
Model fitting time	4 s	4 s	8 s	10 s	~2 min

The best overall accuracy (90.95%) came from CART + Boosting, while the best precision and F1-score for identifying actual subscribers came from C5.0 + CM (70.32% precision, 61.29% F1). Mean 10-fold cross-validation accuracy ranked the models as CART + Boosting > C5.0 + Boosting > C5.0 > CART > C5.0 + CM.

(b) Critical Evaluation

The reported accuracy figures (around 90%) look impressive but are somewhat inflated by the class imbalance in the dataset: because non-subscribers vastly outnumber subscribers, a model can score highly on accuracy while still misclassifying a large share of the minority class, which is exactly what happens with plain C5.0 and CART (sensitivity near 0.63–0.65). The more informative metrics here are precision, sensitivity and F1-score on the positive class, and even the best of these (C5.0 + CM) still leaves roughly 30% of true subscribers unidentified.

Potential sources of bias include: (i) the use of a single historical dataset from one bank and one country, which limits external validity; (ii) no explicit description of how the train/test split was randomised, which could affect reproducibility; and (iii) no confidence intervals or statistical-significance tests around the accuracy differences, several of which (e.g., 90.31% vs 90.95%) are small enough to be within normal run-to-run variation. Additional experiments that would strengthen the findings include repeated stratified cross-validation with variance reporting, benchmarking against non-tree classifiers (logistic regression, random forest, gradient boosting, SVM), and testing on a second, independently collected marketing dataset to confirm the results are not dataset-specific.

(c) Real-World Applicability

The technique is directly deployable in retail-bank marketing operations to rank customers by likelihood of subscribing to a term deposit, allowing call centres to prioritise outreach and reduce wasted contacts — the same approach generalises to other churn or up-sell prediction problems in telecoms, insurance, and retail. Scaling the solution to a production environment introduces several challenges: data availability and quality (customer data must be continuously refreshed and re-validated for drift), compute and latency requirements (the boosted CART model took roughly two minutes to fit, which is manageable offline but would need optimisation for frequent retraining), and ethical/regulatory issues (the input attributes include personal financial and demographic information, so the deployment would need to comply with data-protection regulation and avoid discriminatory targeting based on protected attributes).

Task 4: Reflection & Learning Outcome

(a) Three Key Insights

- Insight 1 — Splitting criteria matter: Seeing C5.0 (information-gain ratio) and CART (Gini coefficient) applied to the same data made concrete how different impurity measures can produce very different tree structures (414 vs 74 decision nodes) from the same 16 attributes, reinforcing the course's coverage of decision-tree induction and attribute-selection criteria.
- Insight 2 — Boosting is not a universal fix: The article shows boosting improving CART noticeably but barely helping C5.0, which deepened my understanding that ensemble/boosting methods (a core CO4 topic) interact with the base learner's own bias-variance profile rather than uniformly improving any model.
- Insight 3 — Evaluation metric choice changes the "best" model: Ranking by accuracy versus by F1-score/sensitivity on the minority class produced different "winning" models, which connects to the course's statistical-learning evaluation concepts (precision, recall, F1) and shows why a single metric can be misleading on imbalanced data.

(b) Proposed Extension

I would propose extending this research by applying the same C5.0/CART + boosting + cost-matrix pipeline to a modern, larger telecom-churn dataset while adding SMOTE-based oversampling of the minority class alongside the existing cost matrix, and benchmarking against a gradient-boosted tree baseline (e.g., XGBoost or LightGBM). This is feasible because the authors' preprocessing pipeline (correlation/VIF screening, confusion-matrix evaluation, k-fold cross-validation) transfers directly to any tabular binary-classification dataset, and open-source implementations of SMOTE and gradient boosting are freely available. The expected impact is a clearer picture of whether the modest gains from simple boosting in the original paper are matched or exceeded by resampling plus modern gradient-boosted trees, which would give practitioners more concrete guidance on which combination of techniques best handles class imbalance in real marketing-analytics settings.

Reference

- Zhao, L., Lee, S., & Jeong, S.-P. (2021). Decision Tree Application to Classification Problems with Boosting Algorithm. *Electronics*, 10(16), 1903. <https://doi.org/10.3390/electronics10161903>
- Rabcan, J., Levashenko, V., Zaitseva, E., Kvassay, M., & Subbotin, S. (2019). Application of Fuzzy Decision Tree for Signal Classification. *IEEE Transactions on Industrial Informatics*, 15(10), 5425–5434. <https://doi.org/10.1109/TII.2019.2904845>
- Sun, R., Wang, G., Zhang, W., Hsu, L.-T., & Ochieng, W. Y. (2020). A gradient boosting decision tree based GPS signal reception classification algorithm. *Applied Soft Computing*, 86, 105942. <https://doi.org/10.1016/j.asoc.2019.105942>
- Yang, Y. (2021). The Evaluation of Online Education Course Performance Using Decision Tree Mining Algorithm. *Complexity*, 2021, Article 5519647. <https://doi.org/10.1155/2021/5519647>
- Alotaibi, A., Shiblee, M., & Alshahrani, A. (2021). Prediction of Severity of COVID-19-Infected Patients Using Machine Learning Techniques. *Computers*, 10(3), 31. <https://doi.org/10.3390/computers10030031>
- An Assessment of Decision Tree based Classification and Regression Algorithms (2020). IEEE Conference Publication. IEEE Xplore. <https://ieeexplore.ieee.org/document/9034296>